

Smart Bin Waste Separation Mechanism

The smart bin identifies each item on a curved receiving plate, aligns the plate opening with the matching waste compartment, then lifts and tilts the plate so the item drops into the selected column. A camera mounted above the plate captures the item, and a microcontroller-based control system coordinates classification and movement.

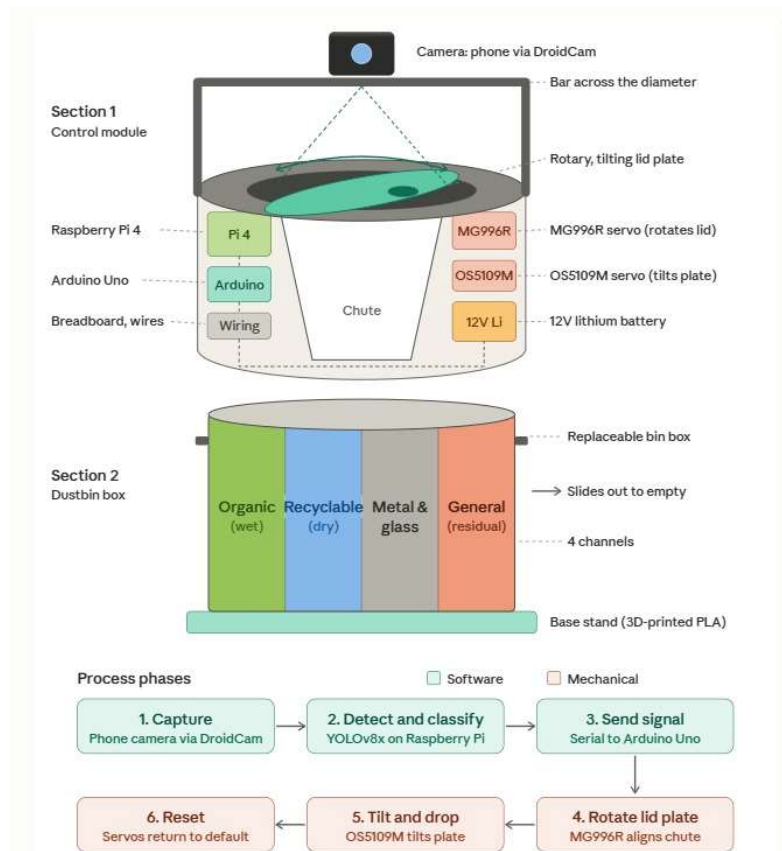


System at a glance

1. Capture	2. Classify	3. Direct and release
Camera records the item placed on the plate.	Vision algorithm assigns a waste category.	Plate rotates to the matching column, then lifts to drop the item.

How the sorting cycle works

- 1. Place the waste:** The user places one item on the curved plate. The plate holds the item above the chute while the system inspects it.
- 2. Capture and classify:** A camera mounted on the bar above the plate captures an image. The vision algorithm running on the microcontroller or connected compute module predicts the waste category.
- 3. Align the outlet:** The controller maps the predicted category to a compartment and commands the rotation motor/servo. The plate turns until its opening is aligned with the selected column.
- 4. Lift and release:** A lifting/tilting actuator raises the plate. The item slides or falls through the aligned opening and chutes into the designated compartment.
- 5. Return to ready:** After the item clears, the actuators return the plate to its receiving position, and the system waits for the next item.



System layout and control sequence reference

Plate rotation and lifting mechanism

The plate assembly combines two motions. First, a rotary stage positions the outlet above the selected waste column. Second, a lifting and tilting linkage opens the plate, so gravity carries the item through the chute. The camera remains fixed above the receiving area to view each item before the plate moves.



Raised curved plate exposing the rotary bearing and tilt linkage beneath it

Control and safety logic

The controller should release the plate only after the category is assigned and the selected outlet is aligned. Position feedback can confirm rotation and lift travel; a timeout or uncertain classification can keep the plate closed and request another scan or

user intervention. These are recommended control behaviors for the concept and should be validated during prototyping.

Problems we might face in future while making it:

1. We might face problems in the lifting mechanism part.
2. Size of curved plate, such that any big waste will not stuck while falling from, also any small waste will not fall early/directly.

References

Wastech, "AI-automated waste sorting pilot," *Wastech Project Details*.

<https://wastech.sa/project-details.html#ai-automated-waste-sorting-pilot>.